

ECE 501: Digital Image Processing

Course Project – Week 3 Progress Report

Adaptive Fabric Surface Defect Detection and Localization using Classical Texture Descriptors

Team Members

Belisha Desai (AU2440034)
Prakshal Shah (AU2440265)
Namyaa Parmar (AU2440025)
Kahaan Shah (AU2440047)

Project Title

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1 Work Done This Week

1.1 Diagnosing the Stage 5 Runtime Bottleneck

Following last week’s dataset characterization, we attempted to run the full anomaly-detection stage (Stage 5) across all 71,120 extracted patches. Combined feature extraction (GLCM + LBP + Gabor + FFT per patch) did not complete in a practical timeframe, since the vast majority of patches (70,645) are normal and far exceed what is needed to fit a stable per-fabric covariance matrix. We profiled this and confirmed feature extraction, not model fitting or scoring, as the bottleneck.

1.2 Redesigning the Anomaly-Model Pipeline for Tractability

We redesigned Stage 5 around two changes: (i) capping the number of normal patches used to train each fabric’s model (rather than using every available normal patch), and (ii) parallelizing feature extraction across CPU cores. This reduces the number of patches requiring feature extraction by roughly an order of magnitude while keeping enough samples to fit a 68-dimensional covariance matrix reliably.

1.3 Addressing the Fabric-Coverage Gap

Our Week 2 audit found defect-free images cover only 20 fabric IDs while defective images span 106, with limited overlap. We implemented an adaptive fallback strategy: a per-fabric Mahalanobis model is used wherever a fabric type has sufficient normal-image data of its own, and a pooled (cross-fabric) Mahalanobis model is used otherwise. This directly operationalizes the “adaptive” aspect of the project’s design in response to a real limitation of the dataset, rather than assuming uniform fabric coverage.

1.4 Evaluation Utilities

We implemented the remaining evaluation machinery: sliding-window anomaly heatmap generation, percentile-based thresholding of the heatmap into a binary predicted defect mask, and IoU, precision, recall, and confusion-matrix computation against ground-truth masks. These are implemented and tested to run end-to-end on a single image; full-dataset evaluation is scheduled for next week (Section 3).

1.5 Literature Review Progress

We finalized our core reference set: the original AFID dataset paper [1], the foundational GLCM [2] and LBP [3] texture-descriptor papers, a Gabor-filter texture-feature reference [4], and a paper benchmarking handcrafted texture-feature extraction specifically on the AITEX dataset [5], which supports our choice of a multi-descriptor approach over a single descriptor.

2 Work To Be Done

- Run the redesigned Stage 5 pipeline end-to-end across all 103 mask-matched defective images (not a small subset) and report aggregate IoU, precision, recall, F1-score, and a confusion matrix.

- Directly compare per-fabric vs. pooled-model detection performance, since both are now implemented.
- Map the raw numeric defect codes (00–08) to AITEX’s named defect categories using the official documentation.
- Ablate the four texture descriptors (GLCM-only vs. +LBP vs. +Gabor vs. +FFT) to identify which contribute most to detection accuracy.
- Finalize and rehearse the mid-semester presentation (10 minutes, tentatively 12th October).

3 Challenges Faced

- Full-dataset feature extraction over 71,120 patches was computationally impractical within a single Kaggle session; this forced a redesign around capped per-fabric sampling and parallelization rather than a straightforward brute-force run.
- The fabric-ID mismatch between normal and defective images (20 vs. 106 distinct IDs) meant a purely per-fabric modeling approach would leave most defective images without any usable normal-texture reference; the pooled fallback model was designed specifically to address this.
- Choosing a thresholding rule for the anomaly heatmap involves a precision/recall trade-off with no single obviously correct answer; we are using a percentile-based threshold on normal-patch scores as a starting point, to be tuned once full-dataset evaluation numbers are available.

References

- [1] J. Silvestre-Blanes, T. Albero-Albero, I. Miralles, R. Pérez-Llorens, and J. Moreno, “A public fabric database for defect detection methods and results,” *Autex Research Journal*, vol. 19, no. 4, pp. 363–374, 2019.
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- [3] T. Ojala, M. Pietikäinen, and T. Mäenpää, “Multiresolution gray-scale and rotation invariant texture classification with local binary patterns,” *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 24, no. 7, pp. 971–987, 2002.
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- [5] A. Ç. Seçkin and M. Seçkin, “Detection of fabric defects with intertwined frame vector feature extraction,” *Alexandria Eng. J.*, vol. 61, no. 4, pp. 2887–2898, 2022.